

1-a) Definition: An Operating System is the system software that manages a computer hardware and software resources and acts as an intermediary between the user and the computer hardware. It controls the execution of programs and provides an environment in which application software can run.

1-b) Three functions:

- * Memory Management: the OS allocates and deallocates memory to programs as they run, and keeps track of which parts of memory are in use.

- * Process Management: the OS creates, schedules, and terminates processes, deciding which process uses the CPU and for how long.

- * File Management: the OS organizes, stores, retrieves, and manages files and directories on storage devices, and controls access to them.

2a) A compiler translates the entire source program into machine code at once, producing a separate executable file before the the program is run.

2b). An interpreter translates and executes the source program line by line, one statement at a time, without producing a separate executable file.

2c) One further difference (partial): A ~~computer~~ compiled program generally runs faster because translation happens beforehand, whereas an interpreted program runs more slowly since translation happens during execution.